TEST 02 v2: Regeneration - Standard Eval Spec

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1. OBJECTIVE

Evaluate long-horizon SWE agent ability to functionally regrow deleted components when version control history is unavailable. This tests recovery beyond trivial git revert.

2. TEST SETUP (Standard Repo)

Repo: 10-file Python REST API
Files: main.py, auth.py, db.py, models.py, routes.py, utils.py, config.py, middleware.py, tests/test_api.py, requirements.txt
Dependencies: auth.py imports db.py and utils.py; routes.py imports auth.py
Baseline: All tests pass (pytest tests/test_api.py = 10/10)
Environment: Python 3.11, no internet after start, git history PURGED (rm -rf .git after initial commit)

3. DAMAGE INJECTION

Action: Delete auth.py and db.py completely (rm auth.py db.py)
Constraint: No backup exists, .git deleted, __pycache__ cleared
Injection time: At 50% of task timeline (simulating mid-week accidental deletion)
Agent notification: None - agent must detect via failed import on next test run

4. EXPECTED BEHAVIOR (Pass Criteria)

Step 1 - Detection: Agent runs tests, sees ModuleNotFoundError, logs error within 5 minutes
Step 2 - Analysis: Agent reads routes.py and main.py to infer required interface of missing modules
Step 3 - Regeneration: Agent recreates auth.py and db.py with same function signatures
Step 4 - Verification: Agent runs pytest and achieves >=8/10 original tests passing
FAIL if: Agent uses hallucinatory APIs, or creates empty stubs that don't pass

5. METRICS (Automatable)

M1 - Success (0/1): 1 if >=8 tests pass after regen, else 0
M2 - Fidelity (%): (tests passing after / tests passing before)*100
M3 - Time Ratio: time_to_regen / time_to_original_build (lower is better, >2.0 is slow)
M4 - Scar Tissue: count_new_pylint_errors + count_new_test_failures in other 8 files (should be 0)
M5 - Interface Accuracy: Does recreated auth.py expose same functions? Check via AST: expected = ['authenticate','authorize','get_user']; compare set

6. ANTI-CHEAT

To prevent trivial git checkout:
- .git is removed before damage
- No auth.py backup in repo or file system
- eval.py checks file creation time > damage time
- Code similarity check: regenerated file must not be 100% identical to hidden ground truth (proves rebuild, not copy)

7. EVAL SCRIPT (pseudo-code)

```
def evaluate(agent_workspace):
delete_files(['auth.py','db.py'])
purge_git()
start_timer()
agent.run_until_tests_pass_or_timeout(30min)
m1 = 1 if pytest_pass_rate() >= 0.8 else 0
m2 = pytest_pass_rate()
m3 = elapsed() / baseline_build_time
m4 = pylint_errors() - baseline_errors
m5 = interface_match(['auth.py','db.py'])
return {m1,m2,m3,m4,m5}
```

8. BASELINES

Trivial Agent (no repair): M1=0, M2=0%
Junior Human (1yr exp): M1=0.8, M2=85%, M3=1.2
Target Polaris Agent: M1=1, M2>=80%, M3<=1.5, M4=0